

[Boundless Potential Driven by Rapid Adaptability and Proactive Learning]

My greatest professional values are the drive to innovate processes for maximum efficiency and a steadfast dedication to continuous learning. During my time in the Natural Language Processing (NLP) laboratory, I proactively mastered complex technical domains such as knowledge graphs, federated learning, and Large Language Models (LLMs), which built a solid foundation for my multimodal AI research. Rather than simply executing assigned tasks, I focus heavily on driving automation across entire workflows. This commitment to efficiency and execution has enabled me to successfully design, develop, and deliver a high volume of projects within compressed timelines.

I maintain a deeply open-minded approach to learning. Believing that the success of medical AI hinges on thorough understanding of clinical domains and close collaboration with frontline clinicians, I have continuously immersed myself in the distinct clinical workflows of various hospital departments to lead projects to success. In Otolaryngology, I built a local LLM-driven SOAP reporting service for inpatients and am currently patenting an AI platform for tinnitus diagnosis and care in collaboration with medical staff. In Cardiology, I developed an uncertainty-aware meta-learning ensemble framework for low-visibility stent segmentation and spearheaded a multi-center validation study to evaluate the frame-selection robustness of AI-QCA, significantly enhancing the reliability of cardiovascular analysis. Furthermore, my cross-departmental agility extends to deep learning hydronephrosis segmentation in partnership with Pediatric Urology, as well as an automated CT/X-ray web VR viewer pipeline to support real-time 3D registration for Orthopedic Surgery. This rapid adaptability ensures that every solution I deliver is precisely optimized for real-world clinical needs.

The synergy between my adaptive clinical knowledge and full-stack engineering has directly translated into substantial operational innovations:

- **Automated Research Pipelines:** Engineered a 'Claw Team'-based Automated Vessel Segmentator framework for angiography, completely streamlining the workflow from environment setup to model testing. This optimized internal operations, compressing new researcher onboarding time from over a month to instant deployment.
- **Clinical Interface Optimization:** Developed an end-to-end medical image automated labeling system that integrates DICOM and complex clinical data into an interactive UI. This reduced radiologic technologists' annotation time by 66%.
- **Rigorous Clinical Validation:** Refusing to settle for raw model performance, I co-authored a multi-center clinical validation study utilizing large-scale datasets to

systematically evaluate the institutional robustness of AI-QCA software, thereby demonstrating strong academic and research capabilities.

Equipped with a sponge-like ability to rapidly absorb diverse medical domains and the technical execution to eliminate operational overhead through workflow automation, I am confident in my ability to actively participate in multifaceted AI projects and contribute significantly to your organization's growth.